Shadman Sakib

ssakib1@umbc.edu • Personal Website • LinkedIn • Google Scholar

RESEARCH PROFILE

Ph.D. researcher in **Information Systems (AI/ML Track)** at the University of Maryland, Baltimore County (UMBC), developing **reliable, deployable, and evidence-grounded multimodal AI for contactless physiological sensing, trustworthy medical reasoning, and autonomous health monitoring**. Research integrates computer vision, video/time-series modeling, vision-language models, signal processing, uncertainty and reliability estimation, and on-device edge inference. Work has progressed from algorithm development to **peer-reviewed publications, field demonstrations, and testbed validation**, including collaborative experimentation with the **DEVCOM Army Research Laboratory (ARL)** and NSF-supported research infrastructure at UMBC.

EDUCATION

Ph.D. in Information Systems (AI/ML Track) — **Jan. 2024 – Present**

University of Maryland, Baltimore County (UMBC) — *Baltimore, Maryland*

- CGPA: **4.00/4.00**; invited to join The Honor Society of Phi Kappa Phi. Research emphasis: multimodal AI, computer vision, physiological sensing, trustworthy AI, medical reasoning, ubiquitous computing, and edge intelligence.

M.Sc. in Information Systems (AI/ML Track) — **Jan. 2024 – Aug. 2025**

University of Maryland, Baltimore County (UMBC) — *Baltimore, Maryland*

- CGPA: **4.00/4.00**. Graduate work in AI for biomedical applications, medical signal processing, machine learning optimization, NLP, data mining, and statistical learning.

B.Sc. in Electrical and Electronic Engineering — **Aug. 2015 – Aug. 2019**

International University of Business Agriculture and Technology (IUBAT) — *Dhaka, Bangladesh*

- CGPA: **3.96/4.00**; **Rank: 1/66 (Top 1%)**.

RESEARCH AND DEVELOPMENT EXPERIENCE

Graduate Research Assistant, Mobile, Pervasive and Sensor Computing (MPSC) Lab — **Jan. 2024 – Present**

University of Maryland, Baltimore County (UMBC) — *Baltimore, Maryland*

- **Project 1 — Reliable Contactless Physiological Sensing from RGB/NIR/IR Video**
  - Develop multimodal vision and signal-processing pipelines for respiratory-rate and physiological monitoring from subtle respiration-induced body motion, with emphasis on robustness to motion, viewpoint, lighting, occlusion, and signal-quality variation.
  - Developed **RespFormer**, a motion-guided temporal-frequency multimodal transformer for respiratory-rate regression and breathing-pattern classification; developed **E2RespUNet**, an attention-enhanced U-Net for end-to-end respiratory waveform reconstruction and rate estimation; optimized inference for NVIDIA Jetson edge hardware.
  - Contributed to **RRPIPS**, formulating respiratory waveform recovery as persistent multi-point motion tracking with coarse-to-fine association and temporal consistency; curated and synchronized multimodal datasets and implemented SQI-based ROI selection, standardized preprocessing, ablation studies, and cross-modal evaluation.
  - Extended this research toward reliability-guided video representation learning, quality-aware sensing, and deployable contactless physiological monitoring under heterogeneous sensing conditions.

  Selected outputs: IEEE ICMLA 2025 • IEEE SmartComp 2025 • IEEE/ACM CHASE 2025 • IEEE/ACM CHASE 2026 • ACM CIKM 2026 **(accepted)** • IEEE/CVF WACV 2027 **(accepted)**
- **Project 2 — Trustworthy Vision-Language and Evidence-Grounded Reasoning for Medical AI**
  - Develop reliability-aware multimodal reasoning methods that explicitly separate sensing and perception quality from downstream reasoning, calibrate available evidence, audit candidate predictions, and reduce unsupported conclusions in physiological and medical AI workflows.

- Developed **VLS-RR**, a perception–reasoning framework for contactless respiratory monitoring; **CALIBRA**, a calibration-aware multi-agent verification framework; and **EviCal-RR**, an evidence-calibrated reasoning framework for reliable physiological inference.
- Investigate active evidence acquisition, selective auditing, counterfactual auditing, grounded evidence verification, and quantitative temporal reasoning in physiological video and multimodal health agents.

Selected outputs: IEEE/ACM CHASE 2026 • CVPR 2026 GRAIL-V • MICCAI 2026 SAFER • AAAI 2026 Fall Symposium **(accepted)** • NeurIPS 2026 Med-Reasoner **(2 papers accepted)**

Current manuscripts: ACM Transactions on Computing for Healthcare

- **Project 3 — Multimodal Foundation Models and Agentic Physiological AI**
  - Build video–time-series and VLM-guided models that align visual, physiological, and contextual streams for contactless monitoring, cross-subject generalization, and low-data adaptation.
  - Investigate masked temporal modeling, multi-scale contrastive alignment, parameter-efficient fine-tuning (PEFT), uncertainty-aware evaluation, fairness across sensing conditions, and efficient edge deployment through quantization and knowledge distillation.
  - Developed **ViTaL-Diff**, a video-token latent diffusion framework for respiratory waveform reconstruction with uncertainty-aware physiological inference.
  - Explore agentic multimodal workflows in which foundation models coordinate perception, evidence assessment, physiological reasoning, verification, and reporting across visual and wearable sensing streams.

Selected outputs: ICML 2026 SD4H

Graduate Research Assistant, Center for Real-time Distributed Sensing and Autonomy (CARDS) **May 2025 – Aug. 2025**

UMBC – Collaborative Research with DEVCOM Army Research Laboratory (ARL) *Baltimore, Maryland*

- **Project — Autonomous Contactless Physiological Sensing on Mobile Robots using DVPG**
  - Developed an **on-device contactless physiological sensing system** for autonomous respiratory-rate estimation on a Boston Dynamics Spot robot, deploying the complete computer-vision and signal-processing pipeline on NVIDIA Jetson Orin.
  - Designed robust respiratory-motion extraction using ROI detection, optical flow, temporal filtering, distance profiling, camera-geometry compensation, and signal-quality assessment under changing viewpoints, outdoor illumination, partial occlusion, and robot-induced motion.
  - Conducted field experimentation at the **DEVCOM Army Research Laboratory Robotics Research Collaboration Campus (R2C2), Graces Quarters, Maryland**, evaluating the system under realistic robotic operating conditions.
  - Contributed to the **2025 ArtIAMAS University Experimentation Event** and Collaborative Remote Robotics Experiment using the Distributed Virtual Proving Ground (DVPG). **ArtIAMAS Field Experiment**
  - Extended the research to heterogeneous mobile robots and multimodal edge sensing for field respiratory monitoring.

Selected outputs: IEEE DCOSS-IoT 2026

Graduate Research Assistant, Quantitative Ultrasound Imaging (QUEST) Lab **Jan. 2023 – Dec. 2023**

University of North Carolina at Charlotte *Charlotte, North Carolina*

- **Project — Deep Learning and Quantitative Ultrasound for Obstetric Imaging**
  - Investigated non-invasive tissue-imaging biomarkers for early disease screening using quantitative ultrasound, signal processing, deep learning, optimization, multimodal image segmentation, acquisition, and registration workflows.

Research Assistant, Advanced Medical Engineering Research Institute (AMERI) **Sep. 2019 – Sep. 2020**

University of Hyogo *Hyogo, Japan*

- **Project — AI-Assisted Surgical Video Understanding and Navigation**
  - Developed deep-learning methods for real-time processing of wearable smart-glass surgical video, including surgical-phase recognition, hand-frame extraction, surgical-tool detection, and context-aware surgical workflow understanding.

Media feature: **AffordSENS Media Feature**

Selected outputs: Joint IEEE SCIS–ISIS 2020

RESEARCH DEMONSTRATIONS AND FIELD VALIDATION

- **NSF TRACE Testbed Workshop and Field Experiment Demonstration**, Center for Research in Use-Inspired Cyber-Physical Systems (CYPRESS), UMBC — Apr. 2026
  Participated in field experimentation and research demonstrations using the NSF TRACE Testbed for autonomous and cyber-physical systems research. **Demo Gallery** | **Workshop Page**
- **Collaborative Remote Robotics Experiment using the Distributed Virtual Proving Ground (DVPG)**, DEVCOM ARL R2C2, Graces Quarters, Maryland — Aug. 2025
  Demonstrated on-device contactless physiological sensing on a Boston Dynamics Spot robot during the ArtIAMAS University Experimentation Event. **Public Event Evidence**

SELECTED PUBLICATIONS

*Full publication record: **Google Scholar**.*

Peer-Reviewed Journal Articles

1. **Sakib, S.**, Hasan, Z., & Roy, N. A State-Of-The-Art Survey of Remote Photoplethysmography for Contactless Health Parameters Sensing. *WIREs Data Mining and Knowledge Discovery*, 15(3), 2025. **Corresponding author.**
2. Shinde, G., Ravi, A., Dey, E., **Sakib, S.**, Rampure, M., & Roy, N. A Survey on Efficient Vision-Language Models. *WIREs Data Mining and Knowledge Discovery*, 2025.
3. Ahmed, I., Routh, B. B., Kohinoor, M. S. R., **Sakib, S.**, Rahman, M. M., & Azzedin, F. Multi-Model Attentional Fusion Ensemble for Accurate Skin Cancer Classification. *IEEE Access*, 2024.
4. Hasib, K. M., Islam, M. R., **Sakib, S.**, Akbar, M. A., Alam, M. S., & Razzak, I. Depression Detection From Social Networks Data Based on Machine Learning and Deep Learning Techniques: An Interrogative Survey. *IEEE Transactions on Computational Social Systems*, 10(4), 1568–1586, 2023.

Selected Conference and Workshop Proceedings

1. **Sakib, S.** & Roy, N. Pixels-to-Phase: Reliability-Aware Latent State-Space Modeling for Contactless Respiratory Estimation. *IEEE/CVF Winter Conference on Applications of Computer Vision* (*WACV 2027*), Buena Vista, Florida, USA, 2027. **Accepted.**
2. **Sakib, S.** & Roy, N. Measure What Matters: Active Evidence Acquisition for Reliable Physiological Reasoning. *NeurIPS 2026 Workshop on Medical Reasoning with Vision Language Foundation Models* (*Med-Reasoner*), Atlanta, Georgia, USA, 2026. **Accepted.**
3. **Sakib, S.** & Roy, N. Can VLMs Keep Time? Probing Quantitative Temporal Reasoning in Physiological Videos. *NeurIPS 2026 Workshop on Medical Reasoning with Vision Language Foundation Models* (*Med-Reasoner*), Atlanta, Georgia, USA, 2026. **Accepted.**
4. Ahmed, I., Sami, A. A., Absar, S., **Sakib, S.**, Rahman, R., Bhuiyan, S. T., Islam, A., & Alam, S. B. Topology-Reliable Retinal Artery–Vein Segmentation via Spatially-Aware Differentiable Graph Pruning. *Workshop on Biomarker-Driven Medical Vision Intelligence* (*BD-MedVis*), *co-located with the 37th British Machine Vision Conference* (*BMVC 2026*), Lancaster, United Kingdom, 2026. **Accepted.** (Oral Presentation)
5. Sami, A. A., Absar, S., **Sakib, S.**, Ahmed, I., & Tushar, Z. H. Do LLM Agents Follow Therapeutic Boundaries? Evaluating Cross-System Drug Substitution Hallucination. *AAAI 2026 Fall Symposium on Agentic and Trustworthy AI for Health and the Global AI-Ready Nursing Workforce*, Arlington, Virginia, USA, 2026. **Accepted.** (Oral Presentation)
6. **Sakib, S.** & Roy, N. Does the Agent Follow the Evidence? Counterfactual Auditing for Trustworthy Multimodal Health Agents. *AAAI 2026 Fall Symposium on Trustworthy Agentic Systems* (*TAS 2026*), Arlington, Virginia, USA, 2026. **Accepted.** (Oral Presentation)
7. **Sakib, S.**, Patel, H., & Roy, N. Reliability-Guided Video Token Learning for Contactless Vital Sign Estimation. *35th ACM International Conference on Information and Knowledge Management* (*CIKM 2026*), Rome, Italy, 2026. **Accepted.**
8. **Sakib, S.** & Roy, N. EviCal-RR: Evidence-Calibrated Reasoning for Contactless Respiratory Monitoring. *MICCAI 2026 Workshop on Stable Adaptation and Faithful Evaluation of Reasoning in Medical Foundation Models* (*SAFER*), Strasbourg, France, 2026.
9. **Sakib, S.**, Shinde, G., & Roy, N. Decoupling Perception and Reasoning for Contactless Respiratory Rate with Vision Language and Small Language Models. *IEEE/ACM CHASE 2026*, Pittsburgh, Pennsylvania, USA, 2026.
10. **Sakib, S.**, Rampure, M., & Roy, N. A Quality-Aware Framework for Contactless Respiratory Monitoring Under Variable Sensing Conditions. *2nd Workshop on Sensing and Computing for Smart and Connected Health* (*SCH*), *co-located with IEEE/ACM CHASE 2026*, Pittsburgh, Pennsylvania, USA, 2026.
11. **Sakib, S.** & Roy, N. ViTaL-Diff: Video-Token Latent Diffusion for Contactless Respiratory Monitoring. *ICML 2026 Workshop on Structured Data for Health* (*SD4H*), Seoul, South Korea, 2026.

12. Rampure, M.[*], **Sakib, S.**[*], Patel, H., Hasan, Z., & Roy, N. Contactless Respiratory Monitoring on Heterogeneous Mobile Robots: A Multimodal Edge-Computing Framework. *8th International Workshop on IoT Applications and Industry 5.0, co-located with IEEE DCOSS-IoT 2026*, Reykjavik, Iceland, 2026. [*]Equal contribution.
13. **Sakib, S.**, Shinde, G., & Roy, N. CALIBRA: Calibration-Aware Multi-Agent Verification for Contactless Physiological Monitoring. *CVPR 2026 Workshop on Grounded Retrieval and Agentic Intelligence for Vision-Language* (*GRAIL-V*), Denver, Colorado, USA, 2026.
14. **Sakib, S.**, Shinde, G., Chugh, S., Anwar, M. S., & Roy, N. RespFormer: A Motion-Guided Temporal-Frequency Multimodal Fusion Transformer for Contactless Respiratory Monitoring. *IEEE ICMLA 2025*, Boca Raton, Florida, USA, 2025.
15. **Sakib, S.**, Shinde, G., Dey, E., & Roy, N. E2RespUNet: End-to-End Respiratory Signal Reconstruction and Rate Prediction Using a Unified Attention-Enhanced U-Net. *IEEE SmartComp 2025*, Cork, Ireland, 2025.
16. Hasan, Z., Ahmed, M., **Sakib, S.**, Chugh, S., Khan, M. A., Faridee, A. Z. M., & Roy, N. RRPIPS: Respiratory Waveform Reconstruction using Persistent Independent Particles Tracking from Video. *IEEE/ACM CHASE 2025*, New York, USA, 2025.
17. **Sakib, S.**, Chowdhury, M. J. U., Hira, M. M. H., Choudhury, T. E., & Das, S. Deep Learning-Based Sentiment Analysis of Social Media and E-Commerce Reviews. *IEEE International Conference on Computing and Machine Intelligence* (*ICMI 2025*), Michigan, USA, 2025.
18. **Sakib, S.**, Siddiky, M. N. A., Arifuzzaman, M., & Chowdhury, M. J. U. Optimizing Facial Recognition: An Analytical Comparison of Traditional and Deep Learning Approaches. *IEEE International Conference on Data Science and Its Applications* (*ICoDSA 2024*), Bali, Indonesia, 2024.
19. Hasib, K. M., **Sakib, S.**, Mahmud, J. A., Mithu, K., & Rahman, M. S. COVID-19 Prediction based on Infected Cases and Deaths of Bangladesh using Deep Transfer Learning. *IEEE World AI IoT Congress* (*AIIoT 2022*), Seattle, USA, 2022.
20. **Sakib, S.**, Hasib, K. M., Tasawar, I. K., Tanzeem, A. K., Arefin, M. F., Islam, S., & Alam, M. S. A Data-Driven Hybrid Optimization Based Deep Network Model for Short-Term Residential Load Forecasting. *IEEE IEMCON 2021*, Vancouver, Canada, 2021. **Best Paper Award.**
21. **Sakib, S.**, Tanzeem, A. K., Tasawar, I. K., Shorna, F., Siddique, M. A. B., & Binte Alam, S. Blood Cancer Recognition Based on Discriminant Gene Expressions: A Comparative Analysis of Optimized Machine Learning Algorithms. *IEEE IEMCON 2021*, Vancouver, Canada, 2021.
22. **Sakib, S.**, Yasmin, N., Tanzeem, A. K., Shorna, F., Hasib, K. M., & Binte Alam, S. Breast Cancer Detection and Classification: A Comparative Analysis Using Machine Learning Algorithms. *3rd International Conference on Communication, Computing and Electronics Systems* (*ICCCES 2021*), Springer, 2021.
23. **Sakib, S.**, Yasmin, N., Tasawar, I. K., Aziz, A., Siddique, M. A. B., & Khan, M. M. R. Performance Analysis of Machine Learning Approaches in Diabetes Prediction. *IEEE Region 10 Humanitarian Technology Conference* (*R10 HTC 2021*), Bangalore, India, 2021. **Best Paper Award.**
24. Shishir, F. S., Hasib, K. M., **Sakib, S.**, Maitra, S., & Shah, F. M. De Novo Drug Property Prediction using Graph Convolutional Neural Networks. *IEEE R10 HTC 2021*, Bangalore, India, 2021.
25. **Sakib, S.**, Siddique, M. A. B., Khan, M. M. R., Yasmin, N., Aziz, A., Chowdhury, M., & Tasawar, I. K. Transfer Learning Based Method for Automatic COVID-19 Cases Detection in Chest X-Ray Images. *IEEE International Conference on Smart Electronics and Communication* (*ICOSEC 2021*), Trichy, India, 2021.
26. **Sakib, S.**, Hossain, B., Hiranaka, T., & Kobashi, S. Hand Frame Extraction in Surgical Video Images Using Convolutional Neural Network. *Joint IEEE SCIS–ISIS*, Hachijo Island, Tokyo, Japan, 2020.
27. **Sakib, S.**, Siddique, M. A. B., & Rahman, M. A. Performance Evaluation of t-SNE and MDS Dimensionality Reduction Techniques with KNN, ENN and SVM Classifiers. *IEEE Region 10 Symposium* (*TENSYMP 2020*), Dhaka, Bangladesh, 2020.
28. Siddique, M. A. B., **Sakib, S.**, Khan, M. M. R., Tanzeem, A. K., Chowdhury, M., & Yasmin, N. Deep Convolutional Neural Networks Model-based Brain Tumor Detection in Brain MRI Images. *IEEE I-SMAC 2020*, Palladam, India, 2020.
29. Khan, M. M. R., **Sakib, S.**, Siddique, M. A. B., Chowdhury, M., Hossain, Z., Aziz, A., & Yasmin, N. Automatic Detection of COVID-19 Disease in Chest X-Ray Images using Deep Neural Networks. *IEEE R10 HTC 2020*, Kuching, Malaysia, 2020.
30. Siddique, M. A. B., **Sakib, S.**, & Rahman, M. A. Performance Analysis of Deep Autoencoder and NCA Dimensionality Reduction Techniques with KNN, ENN and SVM Classifiers. *IEEE International Conference on Innovation in Engineering and Technology* (*ICIET 2019*), Dhaka, Bangladesh, 2019.
31. Siddique, F., **Sakib, S.**, & Siddique, M. A. B. Recognition of Handwritten Digit using Convolutional Neural Network in Python with Tensorflow and Comparison of Performance for Various Hidden Layers. *IEEE International Conference on Advances in Electrical Engineering* (*ICAEE 2019*), Dhaka, Bangladesh, 2019.

32. Khan, M. M. R., **Sakib, S.**, Arif, R. B., & Siddique, M. A. B. Digital Image Restoration in Matlab: A Case Study on Inverse and Wiener Filtering. *International Conference on Innovation in Engineering and Technology (ICIET 2018)*, Dhaka, Bangladesh, 2018.

Selected Manuscripts Under Review

1. **Sakib, S.** & Roy, N. PhyResp: Robust Contactless Respiratory Sensing with Physics-Constrained Predictive State Modeling.
2. Sami, A. A., Chowdhury, M. J. U., Absar, S., Ahmed, I., **Sakib, S.**, Islam, M. R., Azam, S., & Roy, N. Vision-Language Models in Healthcare: A Comprehensive Review from Data to Diagnosis.
3. **Sakib, S.** & Roy, N. SAGE-RR: Selective Auditing with Grounded Evidence for Robust Contactless Respiratory Rate Estimation.

AWARDS, HONORS, AND COMPETITIVE SUPPORT

- **Doctoral Research Poster Award**, UMBC COEIT Research Day — 2025, 2026
- **NSF Student Travel Grant**, IEEE/ACM CHASE 2026 — 2026
- **Student Travel Grant**, Information Systems Department, UMBC – IEEE ICMLA 2025 — 2025
- **Best Paper Award**, IEEE IEMCON — 2021
- **Best Paper Award**, IEEE R10 HTC — 2021
- **Student Travel Grant**, University of Hyogo – IEEE SCIS–ISIS 2020 — 2020
- **HUMAP Fellowship**, University of Hyogo, Japan — 2019–2020
- **Outstanding Chancellor's Teaching Assistant Award**, Department of EEE, IUBAT — 2018–2020
- **University Merit Scholarship** – Undergraduate academic excellence (CGPA ≥ 3.9) — 2016–2019
- **Dean's List of Honors** – all academic levels in undergraduate studies — 2016–2019
- **Chancellor's Award for Academic Excellence** — 2016
- **Departmental Chair's Award** – outstanding undergraduate academic performance — 2015–2016

PEER REVIEW, PROGRAM COMMITTEES, AND PROFESSIONAL SERVICE

- **Journal Reviewer:** Applied Soft Computing; Array; Biomedical Signal Processing and Control; Computational Biology and Chemistry; Computers in Biology and Medicine; Data in Brief; Frontiers in Computer Science; Healthcare Analytics; Informatics in Medicine Unlocked; Intelligence-Based Medicine; Medical Engineering & Physics; Multimedia Tools and Applications; Pulmonary Medicine.
- **Conference and Workshop Reviewer:** IEEE ISBI 2027; WACV Workshop on RWS (2027); ICVGIP (2026); IEEE BigData (2026); MICCAI (2026); AusDM (2026); NCA (2026); IEEE ICSIPA (2026); ICLR Workshop on TSALM (2026); CVPR Workshop on Med-Reasoner (2026); ICML Workshop on FMSD (2026); ICML Workshop on SD4H (2026); ICML Workshop on MusIML (2026); ECCV Workshop on HCV (2026); NeurIPS Workshop on Med-Reasoner (2026); NeurIPS Workshop on JUDGe (2026); NeurIPS Workshop on SocialAgent (2026); NeurIPS Workshop on MusIML (2025); IEEE SCIS–ISIS (2024, 2026).
- **Program Committee and Reviewer**, AAAI-26 Fall Symposium Series (FSS-26) — 2026
- **Organizing Team Member**, NSF TRACE Testbed Workshop Spring 2026, UMBC — Apr. 2026
- **Local Organizer**, IEEE PerCom — 2025
- **Mentor**, IEEE Leading University Student Branch — 2021–2022

TEACHING AND MENTORING EXPERIENCE

Graduate Teaching Assistant, Department of Information Systems — Jan. 2024 – Present

University of Maryland, Baltimore County (UMBC) — *Baltimore, Maryland*

- **IS 420: Database Application Development** – Aug. 2026–Present; Aug.–Dec. 2025; Jan.–May 2025. Led weekly lab sections for **80+ students** covering SQL and PL/SQL, ER modeling, normalization, stored procedures, triggers, and transaction control; designed and graded labs and examinations and evaluated course projects.
- **IS 620: Advanced Database Projects** – Jan.–May 2026; Aug.–Dec. 2024. Supported a project-based graduate course for **50+ students** in advanced database design, schema development, query optimization, and transaction processing; reviewed proposals and evaluated final deliverables.
- **IS 427: Introduction to Artificial Intelligence** – Jan.–May 2024. Assisted **45+ students** with search, probabilistic reasoning, supervised learning, and Python/scikit-learn programming projects.

Lecturer, Department of Computer Science and Engineering **Oct. 2020 – Dec. 2022**

Leading University *Sylhet, Bangladesh*

- Taught Machine Learning, Artificial Intelligence, Digital Signal Processing, Microprocessor and Interfacing, Electrical Circuits, Fundamentals of Computer Applications, and associated laboratory/project courses.
- Designed syllabi, labs, assignments, quizzes, midterms, final examinations, and project assessments; coordinated practical sessions and viva examinations.
- Supervised **undergraduate theses**, resulting in multiple peer-reviewed publications; mentored students in experiment design, scientific writing, ethics, presentation, and camera-ready manuscript preparation.

TECHNICAL EXPERTISE

- **Programming and Scientific Computing:** Python, C/C++, MATLAB, SQL, Bash/Shell; NumPy, Pandas, SciPy, scikit-learn.
- **Deep Learning and Multimodal Foundation Models:** PyTorch, TensorFlow, Keras, Hugging Face Transformers; vision-language models (VLMs), multimodal large language models (MLLMs), video–time-series modeling, multimodal fusion, cross-modal attention, contrastive learning, diffusion models, and foundation-model adaptation.
- **Computer Vision and Medical Imaging:** OpenCV, torchvision, Albumentations, Detectron2; optical flow, motion and point tracking, object detection, segmentation, image registration, ROI extraction, video understanding, and medical/physiological image analysis.
- **Physiological Signal and Time-Series Analysis:** Contactless physiological sensing, respiratory-rate estimation, waveform reconstruction, temporal and spectral signal processing, signal-quality indices (SQI), multimodal sensor synchronization, temporal-frequency modeling, and physiological time-series learning.
- **Trustworthy, Reasoning, and Agentic AI:** Evidence-grounded multimodal reasoning, uncertainty and reliability estimation, calibration, selective auditing, counterfactual evaluation, active evidence acquisition, RAG, agentic AI workflows, and quantitative temporal reasoning.
- **Efficient and Edge AI:** NVIDIA Jetson Nano, Xavier, and Orin; CUDA/cuDNN; on-device inference, PEFT (LoRA/QLoRA), quantization, pruning, knowledge distillation, and resource-efficient model deployment.
- **Robotics and Cyber-Physical Sensing:** Boston Dynamics Spot, distributed/remote robotic experimentation, multimodal robotic sensing, camera-geometry and distance compensation, and edge-based autonomous physiological monitoring.
- **AI Systems and Reproducible Research:** Docker, Git/GitHub, MLflow, FastAPI, Linux; experiment design, ablation studies, cross-modal validation, model evaluation, and reproducible ML pipelines.

PROFESSIONAL MEMBERSHIPS

IEEE Graduate Student Member; ACM Student Member; ACM SIGIR Member; AAAI Student Member.